MICHAEL TRAN

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EDUCATION

Texas A&M University, College Station, Texas **August 2025 - May 2029**

Bachelor of Science in Mechanical Engineering (Engineering Honors), Minor in Mathematics

- **Overall GPA: 4.00**
- Relevant Completed Coursework: Differential Equations, Linear Algebra, Calculus I-III, Newtonian Mechanics, Electricity and Magnetism, Introduction to Statistics, General Chemistry I (Lecture and Lab), Honors Engineering Lab I & II

ENGINEERING PROJECTS

Remote Controlled Trash Collecting Catamaran **June 2026 - July 2026**

- Engineered and manufactured a remote-controlled, 3D-printed catamaran in SolidWorks and Bambu Studio to collect floating debris from lakes and ponds, applying the engineering design process from concept through testing and iteration
- Programmed an ESP32 in C++ (Arduino IDE) and integrated Blynk IoT, dual L298N motor drivers, and differential-thrust propulsion for wireless smartphone control of the boat
- Improved reliability through iterative prototyping; redesigned the hull for better buoyancy, minimized electronics enclosures warping, and sealed critical components against water intrusion

Two-Player Stimulation Clicker **November 2025 - December 2025**

- Collaborated in a team of four to build a two-player version of the online clicker game "Stimulation Clicker" using Python and Pygame, integrating real-time gameplay mechanics and visual feedback with animations and interactive assets
- Devised a round-based scoring system with countdowns, timers, and player turn management for competitive functionality
- Implemented a shop and upgrade system, including boosters, mini-games, and point multipliers, to enhance strategy and replayability

Life-Sized Trebuchet **October 2024 - November 2024**

- Modeled and constructed a 6-foot-tall trebuchet, demonstrating principles of torque and energy transfer
- Conducted tests to ensure stability and launch accuracy, successfully launching a 2-pound projectile over 30 meters
- Gained experience in hands-on prototyping, woodworking, and problem-solving in a team setting

EXPERIENCE

Texas A&M University - College of Engineering | *Peer Teacher (PT)* **August 2026 - December 2026**

- Mentored 30+ first-year engineering students in ENGR 102 (Computation & Python) on core programming concepts, coding logic, and debugging
- Held weekly open office hours, guiding students through lab assignments, computational problem-solving, and code optimization techniques
- Communicated with faculty and other PTs to provide actionable feedback on assignments and reinforce coding principles

Petit Beignets and Tapioca | *Kitchen Helper* **May 2025 - August 2025**

- Prepared and served menu items such as beignets, banh mi, and various finger foods following quality standards
- Handled high-volume periods calmly by prioritizing tasks and communicating effectively as a kitchen team member

ACTIVITIES & VOLUNTEERING

TURTLE Robotics | *Hatchling Member* **January 2026 - May 2026**

- Developed skills in CAD modeling, additive manufacturing (3D printing), C++ programming, circuit design, GitHub utilization, and microcontroller interfacing for robotic systems
- Collaborated with a multidisciplinary student engineering team to map and prototype a semester-long robotics project, aiding in chassis design using SolidWorks
- Utilized the engineering design process including brainstorming, iterative design, prototyping, and results testing

Vertical Flight Society IGNITE Design Challenge | *Competitor* **November 2025**

- Participated in a week-long design challenge, modeling and presenting a winged eVTOL aircraft capable of traversing a series of terrains and delivering a 5-pound payload to a lost hiker using a proximity search algorithm
- Applied design-thinking principles to evaluate feasibility and performance trade-offs
- Worked in a team-based environment to iterate designs and communicate technical ideas clearly

Rhythms of Life Association | *General Member* **August 2022 - May 2025**

- Performed a variety of musical pieces, including both solos and group ensembles at retirement homes for elderly residents

SKILLS

Computer Aided Design: SolidWorks, Bambu Studio
Programming: Python, Java, C++ (learning), Arduino IDE, VS Code
Computer Skills: Excel, Word, Google Workspace